SQL_Queries_Analysis

August 6, 2026

1 SQL Queries Analysis

2 Pizza Sales Analysis Project

2.1 Microsoft SQL Server

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3 Introduction

This document contains all SQL queries used to analyze the Pizza Sales dataset before building the Power BI dashboard.

The SQL queries were written to calculate key business metrics, perform sales analysis, identify customer purchasing patterns, and generate the data required for dashboard visualizations.

The results obtained from these queries were later used to create KPIs and interactive reports in Power BI.

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4 Database Information

Database Name

```sql
CREATE DATABASE pizza_sales;
```

Select Database

```sql
USE pizza_sales;
```

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5 View Dataset

5.1 Business Requirement

Verify that the dataset has been imported successfully.

5.1.1 SQL Query

```sql
SELECT *
FROM pizza_sales;
```

5.1.2 Result

	pizza_id	order_id	pizza_name_id	quantity	order_date	order_time	unit_price	total_price	pizza_size	pizza_category	pizza_ingredients	pizza_name
1	1	1	hawaiian_m	1	2015-01-01	11:38:36.0000000	13.25	13.25	M	Classic	Sliced Ham, Pineapple, Mozzarella Cheese	The Hawaiian Pizza
2	2	2	classic_dlx_m	1	2015-01-01	11:57:40.0000000	16	16	M	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, Ba...	The Classic Deluxe Pizza
3	3	2	five_cheese_l	1	2015-01-01	11:57:40.0000000	18.5	18.5	L	Veggie	Mozzarella Cheese, Provolone Cheese, Smoked Goud...	The Five Cheese Pizza
4	4	2	ital_supr_l	1	2015-01-01	11:57:40.0000000	20.75	20.75	L	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, ...	The Italian Supreme Pizza
5	5	2	mexicana_m	1	2015-01-01	11:57:40.0000000	16	16	M	Veggie	Tomatoes, Red Peppers, Jalapeno Peppers, Red Onio...	The Mexicana Pizza
6	6	2	thai_ckn_l	1	2015-01-01	11:57:40.0000000	20.75	20.75	L	Chicken	Chicken, Pineapple, Tomatoes, Red Peppers, Thai Sw...	The Thai Chicken Pizza
7	7	3	ital_supr_m	1	2015-01-01	12:12:28.0000000	16.5	16.5	M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, ...	The Italian Supreme Pizza
8	8	3	prsc_argla_l	1	2015-01-01	12:12:28.0000000	20.75	20.75	L	Supreme	Prosciutto di San Daniele, Arugula, Mozzarella Cheese	The Prosciutto and Arugula Pizza
9	9	4	ital_supr_m	1	2015-01-01	12:16:31.0000000	16.5	16.5	M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, ...	The Italian Supreme Pizza
10	10	5	ital_supr_m	1	2015-01-01	12:21:30.0000000	16.5	16.5	M	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, ...	The Italian Supreme Pizza
11	11	6	bbq_ckn_s	1	2015-01-01	12:29:36.0000000	12.75	12.75	S	Chicken	Barbecued Chicken, Red Peppers, Green Peppers, To...	The Barbecue Chicken Pizza
12	12	6	the_greek_s	1	2015-01-01	12:29:36.0000000	12	12	S	Classic	Kalamata Olives, Feta Cheese, Tomatoes, Garlic, Beef ...	The Greek Pizza
13	13	7	spinach_supr_s	1	2015-01-01	12:50:37.0000000	12.5	12.5	S	Supreme	Spinach, Red Onions, Pepperoni, Tomatoes, Artichoke...	The Spinach Supreme Pizza
14	14	8	spinach_supr_s	1	2015-01-01	12:51:37.0000000	12.5	12.5	S	Supreme	Spinach, Red Onions, Pepperoni, Tomatoes, Artichoke...	The Spinach Supreme Pizza
15	15	9	classic_dlx_s	1	2015-01-01	12:52:01.0000000	12	12	S	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, Ba...	The Classic Deluxe Pizza
16	16	9	green_garden_s	1	2015-01-01	12:52:01.0000000	12	12	S	Veggie	Spinach, Mushrooms, Tomatoes, Green Olives, Feta C...	The Green Garden Pizza
17	17	9	ital_cpcllo_l	1	2015-01-01	12:52:01.0000000	20.5	20.5	L	Classic	Capocollo, Red Peppers, Tomatoes, Goat Cheese, Gar...	The Italian Capocollo Pizza
18	18	9	ital_supr_l	1	2015-01-01	12:52:01.0000000	20.75	20.75	L	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, ...	The Italian Supreme Pizza

5.1.3 Explanation

Displays all records from the `pizza_sales` table for validation before analysis.

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6 KPI Queries

6.1 Query 1 – Total Revenue

6.1.1 Business Requirement

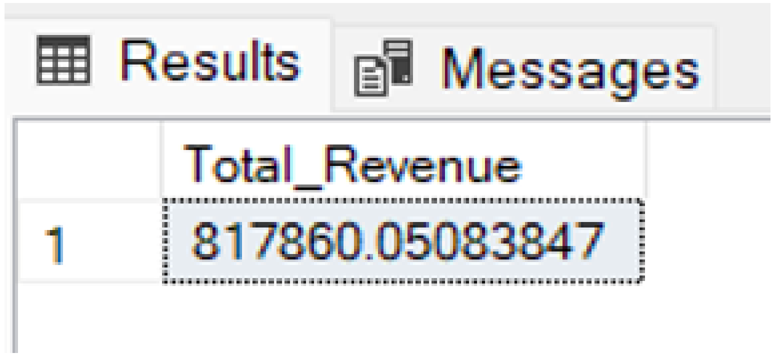
Calculate the total revenue generated from all pizza sales.

6.1.2 SQL Query

```sql
SELECT SUM(total_price) AS Total_Revenue
FROM pizza_sales;
```

6.1.3 Result

Total Revenue: 817.86K



6.1.4 Explanation

Calculates the total revenue by summing the `total_price` column.

6.2 Query 2 – Average Order Value

6.2.1 Business Requirement

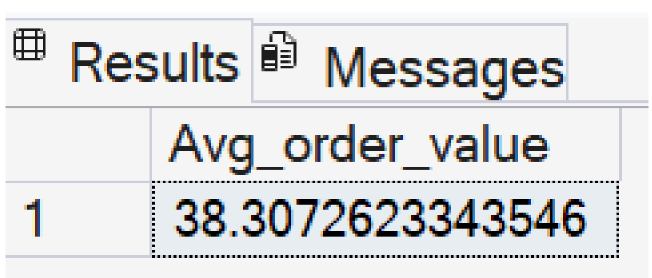
Calculate the average amount spent per customer order.

6.2.2 SQL Query

```
SELECT
SUM(total_price) / COUNT(DISTINCT order_id) AS Avg_Order_Value
FROM pizza_sales;
```

6.2.3 Result

Average Order Value: 38.31



6.2.4 Explanation

Divides total revenue by the total number of unique orders.

6.3 Query 3 – Total Pizzas Sold

6.3.1 Business Requirement

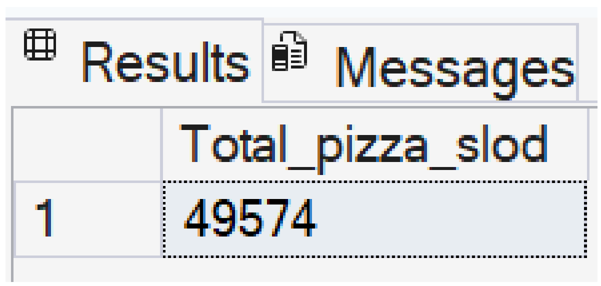
Calculate the total quantity of pizzas sold.

6.3.2 SQL Query

```sql
SELECT SUM(quantity) AS Total_Pizzas_Sold
FROM pizza_sales;
```

6.3.3 Result

Total Pizzas Sold: 49,574



6.3.4 Explanation

Calculates the total number of pizzas sold by summing the `quantity` column.

6.4 Query 4 – Total Orders

6.4.1 Business Requirement

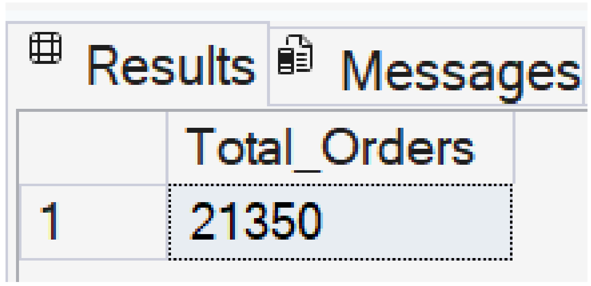
Calculate the total number of customer orders.

6.4.2 SQL Query

```sql
SELECT COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales;
```

6.4.3 Result

Total Orders: 21,350



6.4.4 Explanation

Counts the number of unique orders placed.

6.5 Query 5 – Average Pizzas Per Order

6.5.1 Business Requirement

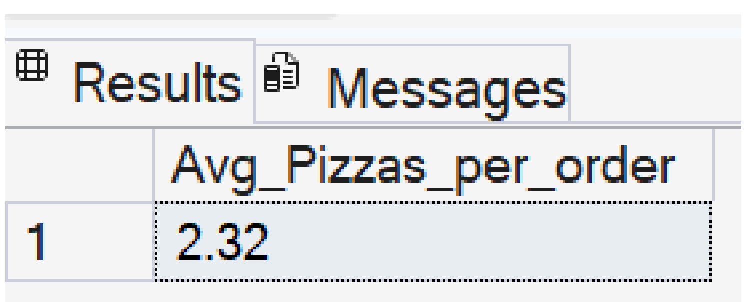
Calculate the average number of pizzas sold per order.

6.5.2 SQL Query

```
SELECT
CAST(
CAST(SUM(quantity) AS DECIMAL(10,2)) /
CAST(COUNT(DISTINCT order_id) AS DECIMAL(10,2))
AS DECIMAL(10,2)
) AS Avg_Pizzas_Per_Order
FROM pizza_sales;
```

6.5.3 Result

Average Pizzas Per Order: 2.32



6.5.4 Explanation

Calculates the average quantity of pizzas purchased in each order.

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7 Dashboard Queries

7.1 Query 6 – Hourly Trend for Total Orders

7.1.1 Business Requirement

Analyze customer ordering patterns throughout the day.

7.1.2 SQL Query

```
SELECT
DATEPART(HOUR, order_time) AS Order_Hour,
COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales
GROUP BY DATEPART(HOUR, order_time)
ORDER BY DATEPART(HOUR, order_time);
```

7.1.3 Result

	order_hours	total_orders
1	9	1
2	10	8
3	11	1231
4	12	2520
5	13	2455
6	14	1472
7	15	1468
8	16	1920
9	17	2336
10	18	2399
11	19	2009
12	20	1642
13	21	1198
14	22	663
15	23	28

7.1.4 Explanation

Groups customer orders by hour to identify peak ordering times.

7.2 Query 7 – Daily Trend for Total Orders

7.2.1 Business Requirement

Analyze customer orders by day of the week.

7.2.2 SQL Query

```sql
SELECT
DATENAME(WEEKDAY, order_date) AS Order_Day,
COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales
GROUP BY DATENAME(WEEKDAY, order_date);
```

7.2.3 Result

Results Messages

	Order_day	Total_Orders
1	Saturday	3158
2	Wednesday	3024
3	Monday	2794
4	Sunday	2624
5	Friday	3538
6	Thursday	3239
7	Tuesday	2973

7.2.4 Explanation

Shows which weekday receives the highest number of customer orders.

7.3 Query 8 – Monthly Trend for Total Orders

7.3.1 Business Requirement

Analyze monthly customer ordering trends.

7.3.2 SQL Query

```
SELECT
DATENAME(MONTH, order_date) AS Month_Name,
COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales
GROUP BY DATENAME(MONTH, order_date);
```

7.3.3 Result

Results Messages

	Month_Name	Total_Orders
1	February	1685
2	June	1773
3	August	1841
4	April	1799
5	May	1853
6	December	1680
7	January	1845
8	September	1661
9	October	1646
10	July	1935
11	November	1792
12	March	1840

7.3.4 Explanation

Summarizes total orders for each month to identify seasonal demand patterns.

7.4 Query 9 – Percentage of Sales by Pizza Category

7.4.1 Business Requirement

Calculate the contribution of each pizza category to total revenue.

7.4.2 SQL Query

```
SELECT
pizza_category,
SUM(total_price) AS Total_Sales,
(SUM(total_price) * 100 /
(SELECT SUM(total_price) FROM pizza_sales)) AS Percentage
FROM pizza_sales
GROUP BY pizza_category;
```

7.4.3 Result

Results Messages

	pizza_category	Total_sales	Percentage
1	Chicken	195919.5	23.9551375322885
2	Supreme	208196.99981308	25.4563112111462
3	Classic	220053.100021362	26.9059602306976
4	Veggie	193690.451004028	23.6825910258677

7.4.4 Explanation

Calculates the percentage contribution of each pizza category toward total sales.

7.5 Query 10 – Percentage of Sales by Pizza Size

7.5.1 Business Requirement

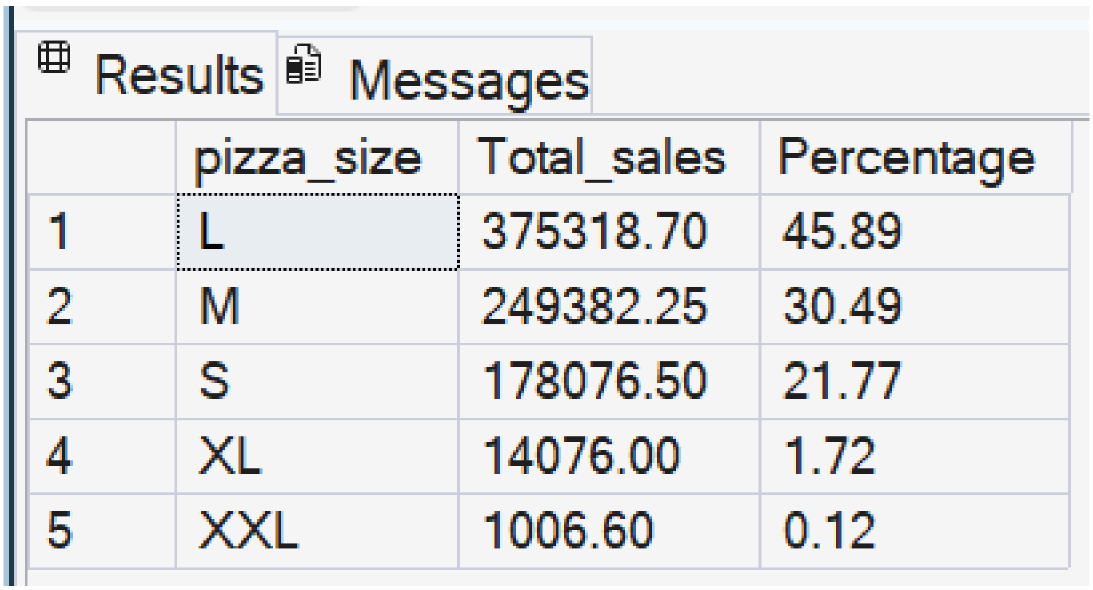
Determine which pizza size contributes the highest revenue.

7.5.2 SQL Query

```
SELECT
pizza_size,
CAST(SUM(total_price) AS DECIMAL(10,2)) AS Total_Sales,
CAST(
(SUM(total_price) * 100 /
```

```
(SELECT SUM(total_price) FROM pizza_sales))
AS DECIMAL(10,2)
) AS Percentage
FROM pizza_sales
GROUP BY pizza_size
ORDER BY Percentage DESC;
```

7.5.3 Result



	pizza_size	Total_sales	Percentage
1	L	375318.70	45.89
2	M	249382.25	30.49
3	S	178076.50	21.77
4	XL	14076.00	1.72
5	XXL	1006.60	0.12

7.5.4 Explanation

Displays revenue contribution for each pizza size.

7.6 Query 11 – Total Pizzas Sold by Category

7.6.1 Business Requirement

Determine which pizza category sells the most units.

7.6.2 SQL Query

```
SELECT
pizza_category,
SUM(quantity) AS Total_Quantity_Sold
FROM pizza_sales
GROUP BY pizza_category
ORDER BY Total_Quantity_Sold DESC;
```

7.6.3 Result

	pizza_category	Total_Quantity_Sold
1	Classic	14888
2	Supreme	11987
3	Veggie	11649
4	Chicken	11050

7.6.4 Explanation

Ranks pizza categories by total quantity sold.

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8 Top & Bottom Seller Analysis

8.1 Query 12 – Top 5 Pizzas by Revenue

```
SELECT TOP 5
pizza_name,
SUM(total_price) AS Total_Revenue
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Revenue DESC;
```

8.1.1 Result

	pizza_name	Total_Revenue
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Pizza	42768
3	The California Chicken Pizza	41409.5
4	The Classic Deluxe Pizza	38180.5
5	The Spicy Italian Pizza	34831.25

8.1.2 Explanation

Returns the five highest revenue-generating pizzas.

8.2 Query 13 – Bottom 5 Pizzas by Revenue

```
SELECT TOP 5
pizza_name,
SUM(total_price) AS Total_Revenue
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Revenue ASC;
```

8.2.1 Result

Results Messages

	pizza_name	Total_Revenue
1	The Brie Carre Pizza	11588.4998130798
2	The Green Garden Pizza	13955.75
3	The Spinach Supreme Pizza	15277.75
4	The Mediterranean Pizza	15360.5
5	The Spinach Pesto Pizza	15596

8.2.2 Explanation

Returns the five lowest revenue-generating pizzas.

8.3 Query 14 – Top 5 Pizzas by Quantity Sold

```
SELECT TOP 5
pizza_name,
SUM(quantity) AS Total_Quantity
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Quantity DESC;
```

8.3.1 Result

Results Messages

	pizza_name	Total_quantity
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

8.3.2 Explanation

Returns the five pizzas with the highest sales quantity.

8.4 Query 15 – Bottom 5 Pizzas by Quantity Sold

```sql
SELECT TOP 5
pizza_name,
SUM(quantity) AS Total_Quantity
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Quantity ASC;
```

8.4.1 Result

Results | Messages

	pizza_name	Total_quantity
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Pizza	950
5	The Soppressata Pizza	961

8.4.2 Explanation

Returns the five pizzas with the lowest sales quantity.

8.5 Query 16 – Top 5 Pizzas by Orders

```sql
SELECT TOP 5
pizza_name,
COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Orders DESC;
```

8.5.1 Result

	pizza_name	Total_orders
1	The Classic Deluxe Pizza	2329
2	The Hawaiian Pizza	2280
3	The Pepperoni Pizza	2278
4	The Barbecue Chicken Pizza	2273
5	The Thai Chicken Pizza	2225

8.5.2 Explanation

Identifies the pizzas included in the highest number of customer orders.

8.6 Query 17 – Bottom 5 Pizzas by Orders

```
SELECT TOP 5
pizza_name,
COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Orders ASC;
```

8.6.1 Result

Results Messages

	pizza_name	Total_orders
1	The Brie Carre Pizza	480
2	The Mediterranean Pizza	912
3	The Spinach Supreme Pizza	918
4	The Calabrese Pizza	918
5	The Chicken Pesto Pizza	938

8.6.2 Explanation

Identifies the pizzas included in the fewest customer orders.

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9 Key Learnings

During this project, the following SQL concepts were applied:

- Database Creation
- Aggregate Functions (`SUM`, `COUNT`)
- Distinct Count
- `GROUP BY`
- `ORDER BY`
- `TOP`
- Date Functions (`DATEPART`, `DATENAME`)
- Arithmetic Calculations
- Percentage Calculations
- Data Aggregation for Business Reporting

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10 Conclusion

SQL played a vital role in extracting meaningful business insights from the Pizza Sales dataset. The queries developed in this project were used to calculate KPIs, analyze sales trends, compare product performance, and prepare data for visualization in Power BI.

This project demonstrates practical SQL skills commonly required in Data Analyst roles, including data aggregation, trend analysis, business reporting, and performance analysis.